

Methanol production from captured CO₂ using hydrogenation and reforming technologies_ environmental and economic evaluation



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ABSTRACT

Minimizing CO₂ emission into the atmosphere to prevent climate change has required immense efforts from us. Flue gases from fossil fuel-based power plants and many heavy industries are considered main anthropogenic sources of releasing CO₂. Although there is a popular solution to mitigate CO₂ through CO₂ capture and storage, its cost is still quite significant, and this prevents broad application. A newly proposed process to convert flue gases into methanol has grabbed a headline because it offers an opportunity to alleviate CO₂ emissions and delivers a way to produce methanol (MeOH) from recycling feedstock. In this work, three approaches for mitigating CO₂ including hydrogenation, bi- and tri-reforming are investigated for methanol production at three different capacities (300, 1500, and 3500 ton/day). The environmental and economic consequences are evaluated as a primary implementation for green process design. These evaluations pointed out that the processes based on reforming are the most appropriate direction for employment during the transition step of producing methanol from a carbon-based- to carbon-free program whereas the hydrogenation-based processes with hydrogen from renewable sources could be the proper implementation scenarios for a long-term plan to obtain near-zero emissions.

1. Introduction

Anthropogenic emissions of CO₂ have surged in recent years as a natural result of social and economic development, resulting in increasing amounts of greenhouse gases in the air. It is important to recognize that this phenomenon has attributed to the rise of global temperature lately which continues to be a problem worldwide. Thus, a rich variety of new strategies is being promoted to mitigate CO₂ emissions by capturing and injecting the CO₂ into the underground [1].

In theory, CO₂ capture and storage (CCS) is regarded as a fruitful method to continue using fossil fuels while lowering CO₂ emissions in the air. In spite of its potential, there are significant weaknesses with regard to CCS options. First, to separate CO₂ from other compounds in the flue gases demands a sizable amount of energy, resulting in a high expenditure of CO₂ treatment [2]. Long-term permanent storages lead to another problem as in the future there might be insufficient storage capacity and/or offshore storages with higher transport and storage costs. In addition, there is the resistance of people to potential carbon storage near inhabited areas. All factors affect the target of an 80 percent emission reduction considerably by 2050 by many countries.

Compared to CCS, CO₂ utilization has shown to be promising in making CO₂ reduction economically attractive. Utilizing CO₂ has

proposed a unique likelihood in reducing emissions by capturing CO₂ from the emission sources, converting it into useful products and yielding revenues. This research aims to use CO₂ as a raw material to synthesize productive products by chemical reactions. In fact, four routes involving thermochemical-, electrochemical-, photo-catalytic-, and bio-enzymatic conversion can be categorized corresponding to raw materials and products and can be used as chemical conversion of CO₂ into diverse products from liquid fuels to inorganic and organic materials [3]. Additionally, synthesizing fuels and chemicals from CO₂ permits one to gradually minimize the subordination to fossil fuels. As a result, the capture and conversion of CO₂ into chemicals appears to present a high potential in terms of environment and economy.

Notwithstanding the mentioned benefits of CO₂ conversion, more attempts are necessary to cope with barriers such as slow kinetics and high energy requirement linked to most CO₂ conversion reactions [4]. These factors result in high economic forfeits and extra CO₂ emissions. Although more studies are conducted with CO₂ conversion reactions at present, the questions related to process development, optimization, economic analysis, and CO₂ reduction have not been answered.

Methanol is an essential intermediate in synthesizing other valuable chemicals, especially dimethyl ether, formaldehyde, MTBE and acetic acid [5]. It is essential to understand that the demand for methanol is

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ever-increasing since it has been widely targeted as a potential candidate to replace gasoline with excellent combustion characteristics and minimal environmental effects. Noticeably, the highest growing need in using methanol and its derivatives has been seen in Asia, particularly in China. A mixture of methanol with conventional gasoline can be used to run in standard engines without any considerable modifications. Additionally, the application of methanol in DMFC (Direct Methanol Fuel Cell) is also seen in electronic devices.

An independent or integrated approach is often utilized to design a methanol plant. While process design and optimization are two key factors in the independent route, the integrated approach allows us to take all stages and units into account. Usually, capital cost and plant operability efficiency are effective tools to evaluate the design and optimization of the conventional approach. Currently, environmental interests have strongly emerged and impacted process concepts. As a result, the integrated approach which combines technological and environmental regulations has gradually become the best choice in the development of a new generation of methanol plants.

Several efforts have been directed to convert CO₂ into methanol through hydrogenation which is a practical process to mitigate the effects of greenhouse gases which is technologically competitive with the conventional methanol process from syngas. Olah and co-workers [5–7] are pioneers in the introduction of alternative pathways for methanol production and its derivatives as well as their use as fuels in the methanol economy. Van-Dal et al. [8] studied methanol production from CO₂ via the hydrogenation route. CO₂ comes from flue gases of a coal power plant while water electrolysis supplied the used hydrogen. Results showed that direct hydrogenation of CO₂ performs remarkably well in abating CO₂ - 1.6 tonCO₂/tonMeOH if O₂ as a by-product is offered for sale or 1.2 tonCO₂/tonMeOH if oxygen is not sold. Machado et al. [9] compared the energy efficiencies and a CO₂ reduction of two methanol processes including the conventional method from syngas and the new method from hydrogenation of CO₂.

Creating syngas by combining CO₂ with natural gas or other hydrocarbons is another way to exploit CO₂ more efficiently. Dry reforming of CO₂ and natural gas generates syngas which has a H₂/CO ratio of 1. However, this ratio is inappropriate for methanol synthesis because producing methanol requires a ratio of around 2. Therefore hydrogen from other sources must be added to syngas to achieve a suitable ratio. In addition, natural gas-based steam reforming usually produces syngas with a ratio of 3, leading to complex separation requirements. To conquer this problem and to create a suitable ratio for methanol production, Olah and co-workers [9] developed a new technology combining steam and dry reforming routes, called “bi-reforming”. Zhang et al. [10,11] studied methanol production from two configurations of the combined reforming of CO₂. The authors used Aspen Plus to construct the flow sheet and simulate the complete process including syngas production and methanol synthesis (excluding methanol purification). The research pointed out that two routes are likely to have economic potential as calculated by the NPV, IRR, and DPBP factors.

Song et al. [12] developed a new process, called “tri-reforming”, a three-step reaction process which has the potential to produce an appropriate H₂/CO ratio for producing different valuable chemicals. Tri-reforming is a synergetic integration of endothermic steam and dry

reforming, and exothermic partial oxidation. The use of different reforming technologies could create syngas with the proper H₂/CO ratio and alleviate the carbon formation. The integration of reforming and partial oxidation technologies allows the process to significantly diminish or get rid of carbon formation on catalysts, bringing on the longer catalyst life and higher process productivity. O₂ plays an essential role in raising energy efficiency by generating heat in situ as well as lowering carbon formation in catalysts. Zhang et al. [13] also had an interest in converting CO₂ of flue stack gases into methanol through the tri-reforming process. With the help of a simulation tool, the authors studied the operating conditions of temperature, pressure, and the feed ratio in obtaining the target ratio of syngas and maximum CO₂ conversion. Cañete B. et al. [14] aimed to produce methanol from natural gas with high CO₂ content through reforming reactions. The study proved that reforming technologies are competitive in comparison with steam reforming in terms of economy.

In this paper, three detailed processes of CO₂ conversion into methanol, namely hydrogenation, bi- and tri-reforming processed were presented. The objectives of this study are (a) to simulate, optimize, and heat integrate conversion processes from CO₂ at three different capacities (300, 1500, and 3500 ton/day), (b) to compare the investment and operational costs, and (c) to evaluate the CO₂ emission reduction.

2. Process parameters

Generally, flue gases are made up of CO₂, N₂, O₂, H₂O, CO, H₂, and several trace components of NO_x, SO_x, and other compounds. Nevertheless, in this study, pure CO₂ captured from flue gases by absorption or other technologies is employed to feed to methanol production processes. The details of the capturing process for CO₂ from different flue gas sources are shown by Tuan et al. [15]. The captured CO₂ needs to have the compositions of at least 95 mol% of CO₂, not more than 4 mol% of N₂ and the remaining to meet the pipeline quality specifications [16]. Also, in this study, natural gas, biomass, and other renewable sources are the candidates to produce hydrogen. Moreover, diverse technologies can be employed to produce hydrogen, including thermal (natural gas reforming, and biomass gasification), electrolytic (water breaking) and photolytic (breaking water by solar energy). With these technologies, very pure hydrogen of 99.99 wt% can be obtained [17]. The hydrogen in this research is purchased from the external sources. The natural gas used in all processes is imported from the natural gas pipeline system and has a composition of 88.71 mol% CH₄, 6.93 mol% C₂H₆, 1.25 mol% C₃H₈, 0.28 mol% n-C₄H₁₀, 0.05 mol% n-C₅H₁₂, 0.02 mol% n-C₆H₁₄, 0.82 mol% N₂, and 1.94 mol% CO₂ [18]. Oxygen could be manufactured from the cryogenic air-distillation process, or the water electrolysis (with electricity from renewable sources) which can generate oxygen with high purity of 99.2–99.99 mol % [19]. In this study, pure oxygen is purchased from external suppliers. Finally, the processes to convert CO₂ into methanol are modeled, simulated, heat integrated, and optimized over three distinct methanol capacities (300, 1500, and 3500 ton/day), corresponding to small-, medium- and large-scale plants. Additionally, the capital, operational and total costs for each scenario are calculated and compared. The conceptual structures for CO₂-based technologies in this study are presented in Figs. 1 and 2.

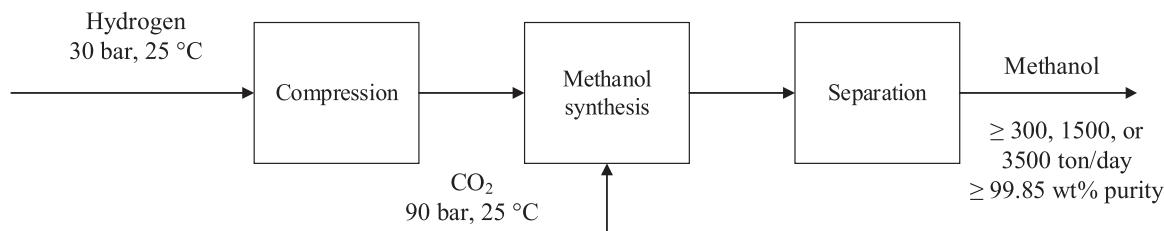


Fig. 1. Schematic diagram of the process based on hydrogenation.

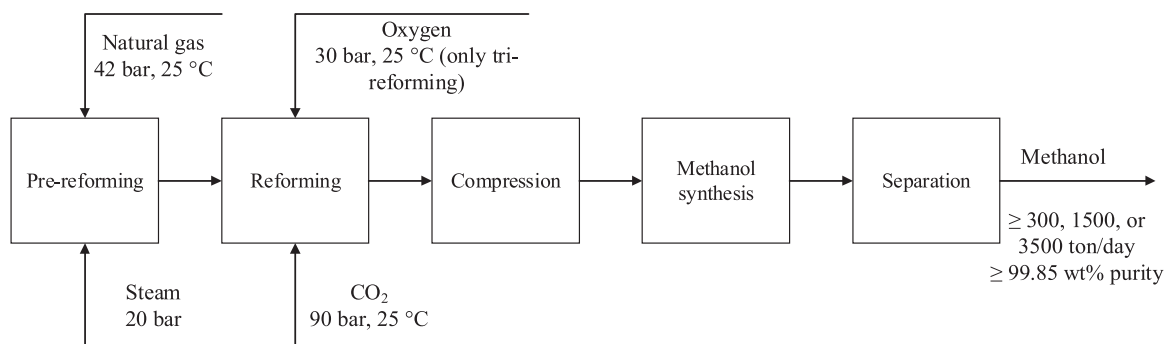


Fig. 2. Schematic diagram of the process based on bi- and tri-reforming.

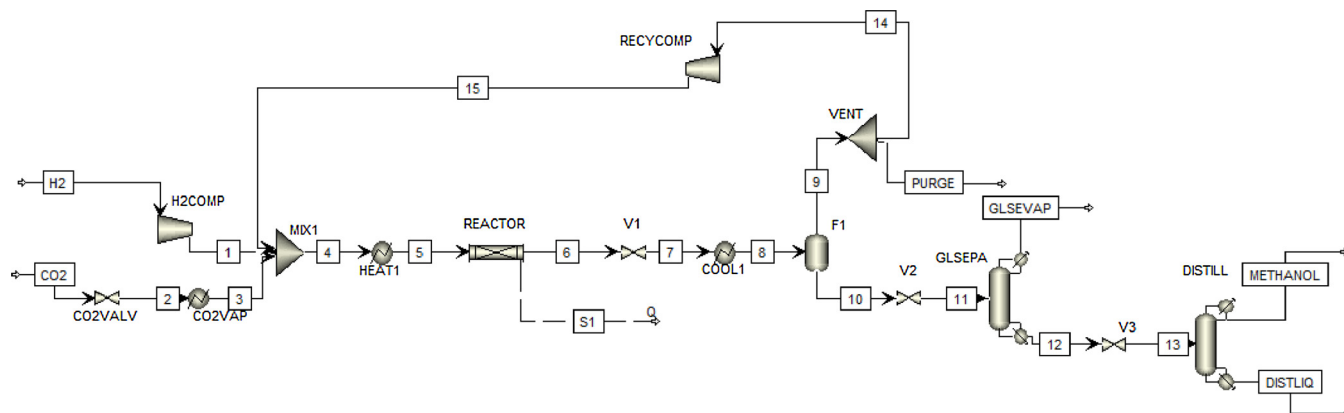


Fig. 3. Flow sheet of the hydrogenation-based processes.

3. Process description and modeling framework

3.1. Hydrogenation-based methanol process

3.1.1. Process overview

The hydrogenation based methanol process is displayed in Fig. 3. The hydrogenation route contains three sections: Methanol synthesis, Separation and Purification, and Heat recovery.

CO₂ at 90 bar and 25 °C which is assumed to originate from the CO₂ pipeline systems [20,21] is decompressed to 50 bar before it is mixed with hydrogen which is then compressed from 30 bar to 50 bar in a single compressor. After mixing, two gas streams are combined with the recycle stream, preheated to 240 °C, and fed to the fixed bed reactor. The outflow is expanded and cooled to 35 °C before injected into the flash drum in which water and methanol are condensed and removed from the residual gases. A part of the vapor flow (1 mol%) is released to the air to evade the buildup of inert gases and by-products in the process while the rest is fed back to the reactor. The flow from the bottom of the flash tank is expanded to 12 bar through valves before injected into the G–L separator. Remaining unreacted gases and crude methanol are separated at the top and bottom of the separator. After expansion to 1.5 bar, the crude methanol is purified to 99.85 wt% at the top of the distillation tower while the bottom product is almost water.

3.1.2. Modeling framework

The flow sheet model of the hydrogenation process was developed in Aspen Plus. RKSMHV2 was utilized to evaluate thermodynamic properties for the high-pressure streams (> 10 bar) while NRTL-RK was applied for other streams.

During the hydrogenation of CO₂ into methanol, these following reactions occur, including hydrogenation of CO₂ and CO, and reverse water gas shift. However, only two of these reactions are linearly independent.



In this study, the isothermal reactor is loaded with a Cu/ZnO/Al₂O₃ commercial catalyst with the model suggested by Van den Bussche and Froment [22], modified by Mignard and Pritchard [23]. Although this model is based on experimental work carried out with CO – CO₂ - H₂ mixtures with low CO₂ content, it predicts good results for the methanol reactions with high CO₂ composition as shown by Van-Dal et al. [8], and Luyben [24]. The reaction equations of this model are displayed in Eqs. (1) – (3). The kinetic model is given by Equations A.1 - A.5 while the kinetic parameters are presented in Table A1.

In this research, the hydrogenation reactor is simulated in Aspen Plus by employing the RPLUG model with a constant thermal fluid temperature. The thermal fluid temperature is controlled at 264 °C, resulting in the production of high-pressure steam (254 °C and 42 bar). The pressure drop across the fixed bed reactor is evaluated by the Ergun equation in Aspen Plus. The catalysts and dimensions of the hydrogenation reactor are listed in Table S1.

A series of distillation towers are employed to reach the desired purity of methanol of 99.85 wt% (i.e., Grade AA). These columns are simulated by the RadFrac model in Equilibrium mode in Aspen Plus. The mass fraction of methanol in the top and bottom of these columns are managed by a reflux ratio and a distillate to feed ratio which are assigned as two design specifications in both columns. The column diameter is designed to perform at 70% of its fractional capacity while the column height is calculated by a number of theoretical plates.

3.2. Bi-reforming-based methanol process

3.2.1. Process overview

The bi-reforming based methanol process is shown in Fig. 4. The bi-

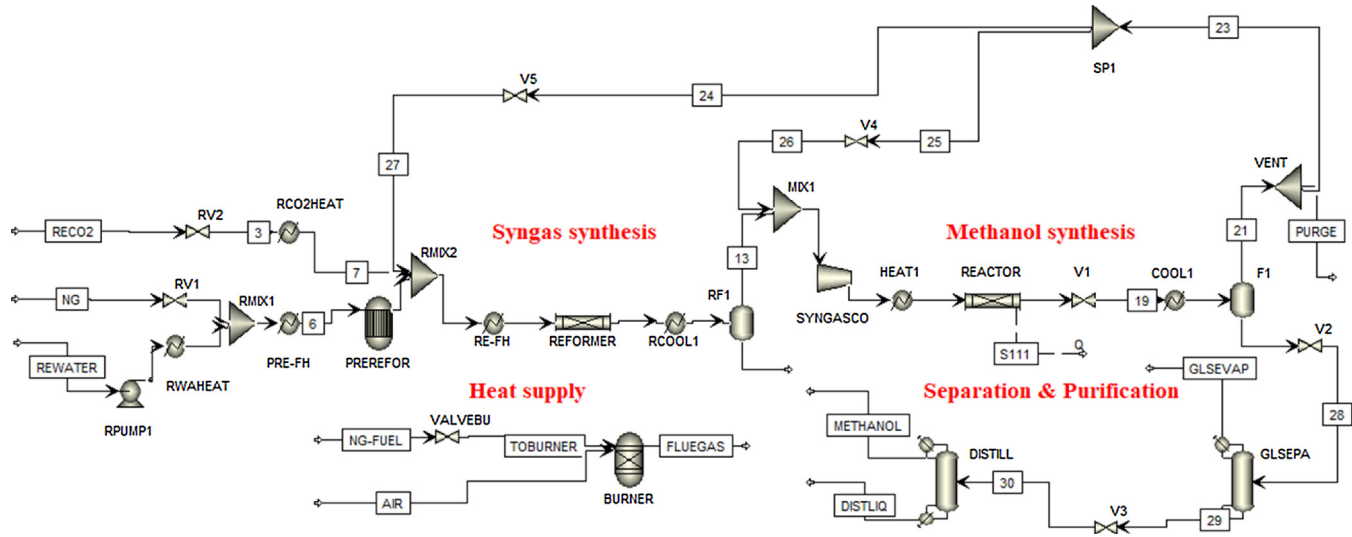


Fig. 4. Flow sheet of the bi-reforming-based processes.

reforming route contains five sections: Syngas synthesis (by bi-reforming), Methanol synthesis, Separation and Purification, Heat supply, and Heat recovery.

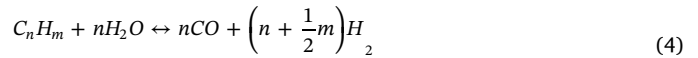
Pipeline-quality natural gas at 42 bar and 25 °C is mixed with high-pressure steam at 20 bar before preheated to 400 °C and entered to the pre-reformer. The pre-reformer output stream then is combined at 20 bar with the recycle-to-reformer stream and CO₂ which is decompressed from 90 bar and 25 °C from the CO₂ pipeline systems. After preheating to 600 °C, the mixing stream enters the reformer. The effluent from the reactor is cooled to 50 °C, which allows water to be condensed and removed from the syngas in the first flash drum. It is noted that syngas is presented by the stoichiometric ratio (SR), $SR = (H_2 - CO_2)/(CO + CO_2)$, with a proper value for methanol production fixed in the range of 2.0 to 2.1 in this study [25]. Syngas from the top of the first flash tank is mixed with the recycle-to-methanol-reactor stream and pressurized to 50 bar. The mixed stream is further heated to 240 °C before entered to the methanol reactor. The outflow is expanded and cooled to 35 °C and injected into the second flash drum to remove most residual gases. A portion of the unreacted gases (1 mol%) is vented to suppress the buildup of inert gases in the process while the rest is fed back to the reformer and the methanol reactor. As shown by Bae [26], the recycle to both the reformer as well as the methanol reactor is likely to improve energy efficiency and carbon availability in the overall process. The flow from the bottom of the second flash tank is then expanded to 12 bar and enters the G–L separator. In this column, unreacted gases and crude methanol are separated into the overhead and bottom sections. At a second column, methanol of 99.85 wt% is obtained at the top while the bottom product contains almost pure water.

3.2.2. Modeling framework

The flow sheet of the bi-reforming process was modeled in Aspen Plus. SRKKD was utilized to evaluate the thermodynamic properties for all streams in the syngas synthesis section. For the methanol synthesis section, RKSMHV2 was utilized to assess the thermodynamic properties for the high-pressure streams (> 10 bar) while NRTL-RK was used for the other streams. This procedure is also used for the tri-reforming process which is presented in the next section.

Steam reforming of higher hydrocarbons could result in carbon formation on the catalyst; therefore, if higher hydrocarbons exist in the natural gas, a pre-reformer is a necessary process step to directly convert them into C1-components (CH₄ and CO₂) with no intermediate products as shown by Eqs. (4)–(6). The pre-reformer is normally operated adiabatically at 400–550 °C using a nickel-containing steam reforming catalyst. In addition, the use of pre-reformer allows decreasing

the size of the reformer as the result of reducing energy from the reformer [27]. Here, the Gibbs reactor is applied to model the pre-reformer under adiabatic conditions. Moreover, under normal operating conditions, the three reactions mentioned in Eqs. (4)–(6) can be assumed to achieve chemical equilibrium, i.e. the reaction rates are very fast at given temperature and pressure.



During the bi-reforming of CO₂ into methanol, the following reactions occur, including steam reforming, dry-reforming, and the water gas shift. However, only two of these reactions are linearly independent from a thermodynamic viewpoint.



In this research, a kinetic model for the combined reforming by a Ni-CeO₂/MgAl₂O₄ catalyst [28,29] is employed to predict the results for the syngas synthesis from bi-reforming reactions. The kinetic equations of this model are given by Equations B.1 - B.10 while the kinetic parameters are presented in Table B1.

In this work, the bi-reformer is modeled in two parts: the bi-reforming reactions by the RPLUG model with a constant at specified reactor temperature at 920 °C and the combustion of fuel (natural gas) and air in the firebox by the Gibbs reactor. An amount of air is provided to give a 10% higher boost of oxygen than the stoichiometric amount required for completely combusting natural gas. The heat discharged from the firebox has to match the heat needed in the reactor. The pressure drop across the reactor is evaluated by the Ergun equation in Aspen Plus. The catalysts and dimensions of the bi-reformer are listed in Table S2.

Similar to the hydrogenation process, the kinetic model for methanol synthesis of the bi-reforming process (also the tri-reforming process, presented in the next section) is based on the model by Van den Bussche and Froment, modified by Mignard and Pritchard. This model delivers good predictions for methanol production from syngas. The methanol reactor is also simulated by the RPLUG model with a constant

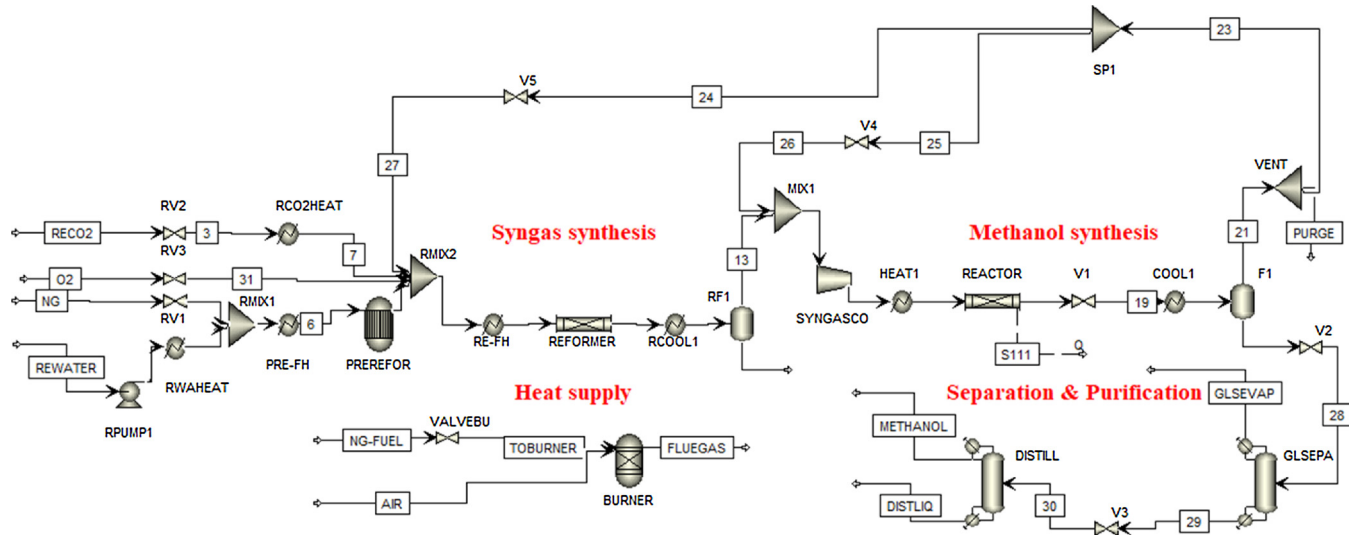


Fig. 5. Flow sheet of the tri-reforming-based processes.

thermal fluid temperature with the Ergun equation for pressure drop calculation. In addition, characteristics of the methanol reactor are presented in Table S1. Furthermore, the bi-reforming process (also the tri-reforming process, presented in the next section) requires two distillation columns which are modeled by the RadFrac model in Equilibrium mode to attain the desired purity of 99.85 wt% at the top of the last column.

3.3. Tri-reforming-based methanol process

3.3.1. Process overview

The tri-reforming based methanol process is described in Fig. 5. Similar to the bi-reforming route, there are five sections in the tri-reforming route: Syngas synthesis (by tri-reforming), Methanol synthesis, Separation and Purification, Heat supply, and Heat recovery.

Pipeline-quality natural gas at 42 bar and 25 °C is decompressed to 20 bar before mixed with high-pressure steam. The mixing stream enters a pre-reformer after preheated to 400 °C. The outflow is then combined with CO₂, oxygen and the recycle-to-reformer stream at 20 bar, preheated to 600 °C before fed to the tri-reformer. The product from the tri-reformer enters the first flash tank to separate condensed water from syngas. Similar to the bi-reforming route, SR of syngas in this case is controlled in the range of 2.0 to 2.1. Next, the syngas is mixed with the recycle-to-methanol-reactor stream and increased to 50 bar and 240 °C before entered to the methanol reactor. After cooled to 35 °C, gas and liquid streams are separated in a second flash tank. While 1%mol of the unreacted gases are purged when the remaining is sent back to the tri-reformer and the methanol reactor. The methanol purity of 99.85 wt% is obtained from the liquid flow of the second flash tank through the usage of a series of two distillation columns.

3.3.2. Modeling framework

Similar to the bi-reforming, the Gibbs reactor for adiabatic conditions is used to model the pre-reformer in this process.

During the tri-reforming of CO₂ into methanol, these following reactions occur, including steam reforming, dry reforming, partial oxidation, and the water gas shift.



However, to describe the tri-reforming process, a series of reactions as described by Eqs. (15)–(18) is considered. These equations can be used to characterize the possible reactions existing in the tri-reforming process as a result of the extensive information accessible in the literature [30].



In this work, an integration of the kinetic model developed by Xu and Froment [31] as well as Trimm and Lam [32] over a Ni-based catalyst are used to present the reaction Eqs. (15)–(18). The kinetic model equations of this model are shown by Equations C.1 - C.9 while the kinetic parameters are shown in Table C1.

Similar to the bi-reforming, the tri-reformer is simulated in two parts: the tri-reforming reactions by the RPLUG model at 920 °C and the firebox by the Gibbs reactor with the Ergun equation for pressure drop calculation. The catalysts and dimensions of the tri-reformer are listed in Table S3.

4. Process economic evaluation and optimization

To guarantee the fulfillment of CO₂-based processes, the following conversion constraints are established:

$$\text{Methanol purity} \geq 99.85 \text{ wt\%} \quad (19)$$

where the Methanol purity is the mass fraction of methanol in the product stream ('Methanol' stream).

$$\text{Methanol capacity} \geq 300, 1500, \text{ or } 3500 \text{ ton/day} \quad (20)$$

where the Methanol capacity is the mass flow rate of the product stream ('Methanol' stream).

The suggested optimization problem can be explained as follows: Given the flow sheet structure displayed in Figs. 3–5, and the feed gas stream conditions, the target is to identify the optimal operating conditions and process synthesis configurations to minimize the annualized total cost while achieving the methanol capacity (300, 1500, or 3500 ton/day) and purity to be at least 99.85 wt%.

$$\min \text{Annualized total cost} = \text{OPEX} + \text{CAPEX} \left(\frac{i(i+1)^N}{(1+i)^N - 1} \right) \quad (21)$$

The annualized total cost is calculated as the sum of the annualized capital cost and the operating cost of the plant (OPEX) as displayed in Eq. (21). The interest rate (*i*) is 8%, and a 30 year lifetime (*N*) was presumed for each plant [33].

The capital cost (CAPEX) is the total cost of the plant which includes direct and indirect costs such as total installed cost, contracts, contingencies, overheads, and other costs. The cost of equipment is evaluated by utilizing the Aspen Economic Analyzer with the costing template for Europe, which allows the on-the-fly sizing and cost estimation of equipment based on an up-to-date data bank [34]. The Chemical Engineering Plant Cost Index (February 2017) is exerted to update the costs of equipment. The operational cost (OPEX) is calculated from the total cost of raw material, utility, operating labor, maintenance, operating charges, plant overhead, and general and administrative expenses by Aspen Economic Analyzer as well. Each plant is expected to operate 8000 h per year. Tables S4 - S5 summarize the utilities and raw prices employed in evaluating OPEX, and Table S6 displays the operational costs assumptions in the study.

The primary decision variables for the hydrogenation-based technology are: a) operating temperature and pressure of the methanol reactor, b) dimensions and a number of tubes of the methanol reactor, c) reflux ratio and distillate to feed ratio of two columns, d) outlet pressure of valves, and e) flash drum temperature. For the bi or tri-reforming-based process, those variables include a) operating temperature and pressure of the reformer and methanol reactor, b) dimensions and number of tubes of the reformer and methanol reactor, c) reflux ratio and distillate to feed ratio of two columns, d) outlet pressure of valves, and e) flash drum temperature. For variables, a local sensitivity analysis is performed to discover the optimum values that display the minimum annualized total cost.

5. Energy network optimization

The next step is to match the hot and cold utility load and to designate the heat exchanger matches, the areas of each match, and the HEN (Heat Exchanger Network) topology [35], that produces the minimum annualized cost. The heat exchanger network design is automatically carried out by utilizing the Aspen Energy Analyzer with Automatic Recommend Designs (ARD) feature. This framework has three steps including: (1) elimination of very heat exchanger poor matches, (2) estimation of optimum heat load distribution, (3) structural optimization that satisfied optimum heat load distribution, which are reported in detail by Tuan et al. [15].

It is essential to show that this approach is carried out for every CO₂-based processes. After attaining the heat integration network, the rigorous simulation with the heat integration result input is executed for each scenario.

6. Waste heat recovery and electricity generation

The heat recovery system is necessary to increase the overall energy efficiency and reduce fuel usage through three different strategies exerted in this study. Firstly, unreacted gases from the purge stream and the top of G-L separator are mixed with air and then combusted to produce high-pressure steam (42 bar). To model this burner, the Gibbs reactor model was utilized, and the operating conditions are 1.2 bar and 1200 °C. Secondly, heat exchanger units with the hot flue gas from burning natural gas for heat supply to the reformer are applied to preheat the feed streams, and generate the steam for bi- or tri-reforming reactions as well as electricity generation. Lastly, to maintain the stability of temperature inside the methanol reactor, the heat from the methanol synthesis reactions has to be removed via a cooling jacket. The residual heat from the cooling system is also used to yield high-pressure steam (42 bar) for further electricity production.

Finally, the annualized total cost could be worked out anew to accomplish the production cost (€/tonMeOH) for each case.

7. Environmental analysis

As CO₂-based processes should reduce CO₂ emissions in the atmosphere, the net CO₂ emission must be evaluated. CO₂ is released from the process in three distinct routes, including (a) CO₂ from flue gases, (b) CO₂ generated by steam production for the heat supply to the process, (c) CO₂ emitted as a result of electricity generation, (d) CO₂ emission associated with H₂, O₂ production as well as the natural gas supply chain. The first is generally called a direct CO₂ emission while the others are referred to as indirect emissions. The value of CO₂ emission as a result of using thermal energy and electricity is 205.3 lbCO₂/MMBtu and 0.596 tonCO₂/MWhr respectively [36]. Moreover, CO₂ emissions as a consequence of producing H₂ or O₂ from different technologies as well as acquiring natural gas from the natural gas supply chain are shown in Table S5. The net CO₂ emission is evaluated by the following equation. This value could be negative when the amount of CO₂ used in the process is higher than that released from the process.

$$n_{CO_2, net} = n_{CO_2, fluegas} + n_{CO_2, indirect} - n_{CO_2, feed} \quad (22)$$

8. Results and discussion

8.1. Process performance results

The CO₂-based processes are modeled, simulated, heat integrated, and optimized over three different methanol capacities (300, 1500, and 3500 ton/day), corresponding to small-, medium- and large-scale plants. In detail, the optimization results are shown in Tables S7 - S9 while Tables S10 - S12 summarize the mass balance of all components entering and exiting the system for three technologies.

For the hydrogenation technology, the introduced amount of H₂ was in excess of the amount necessary for CO₂ hydrogenation (H₂/CO₂ = 3) in all cases. Specifically, 2.64 ton/h of H₂ and 19.69 ton/h of CO₂ are required to produce 300 ton/day of methanol. Meanwhile, the process is fed with 98.43 ton/h of CO₂ and 13.18 ton/h of H₂ to create 1500 tonMeOH/day while the 3500 tonMeOH/day plant orders 229.62 and 30.74 ton/h for CO₂ and H₂ respectively.

For the bi-reforming technology, methanol synthesis gas is normally characterized in terms of the stoichiometric ratio (SR), SR = (H₂ - CO₂)/(CO + CO₂). In theory, the optimum for this ratio is 2. However, in this study the process operates with a few percent excess of hydrogen to control the formation of by-products (SR = 2.05). In addition, to suppress carbon formation requires a suitable steam/carbon ratio. To obtain the target of methanol production rate while SR equal to 2.05, the inlet streams of CO₂, natural gas, and steam are adjusted to 4.57, 5.51, and 13.67 ton/h respectively for 300 tonMeOH/day. These values are 22.82, 27.55, and 68.29 ton/h for 1500 tonMeOH/day while 3500 tonMeOH/day capacity has the demand of 52.70, 64.55, and 140.82 ton/h for CO₂, natural gas, and steam respectively.

For the tri-reforming technology, the syngas SR ratio of 2.05 and a suitable steam/carbon ratio are also required. Moreover, the O₂/CO₂ ratio is set at 0.21 for good conversion of CO₂ [12]. The feed flow rates of CO₂, natural gas, steam, and O₂ are fixed as 3.95, 5.7, 9.61, and 0.61 ton/h respectively for 300 tonMeOH/day. The corresponding values for 1500 tonMeOH/day are 19.72, 28.48, 48.02, and 3.06 ton/h while those for 3500 tonMeOH/day are 46, 66.44, 112.01, and 7.13 ton/h for CO₂, natural gas, steam, and O₂ respectively.

8.2. Economic evaluation

The costs acquired for the processes to convert CO₂ into methanol by utilizing various technologies (hydrogenation, bi- and tri-reforming) with different capacities (300, 1500, and 3500 ton/day) are shown in Figs. 6–9. At this stage of the design process, the cost estimates are still very rough and can vary up till 40% [37].

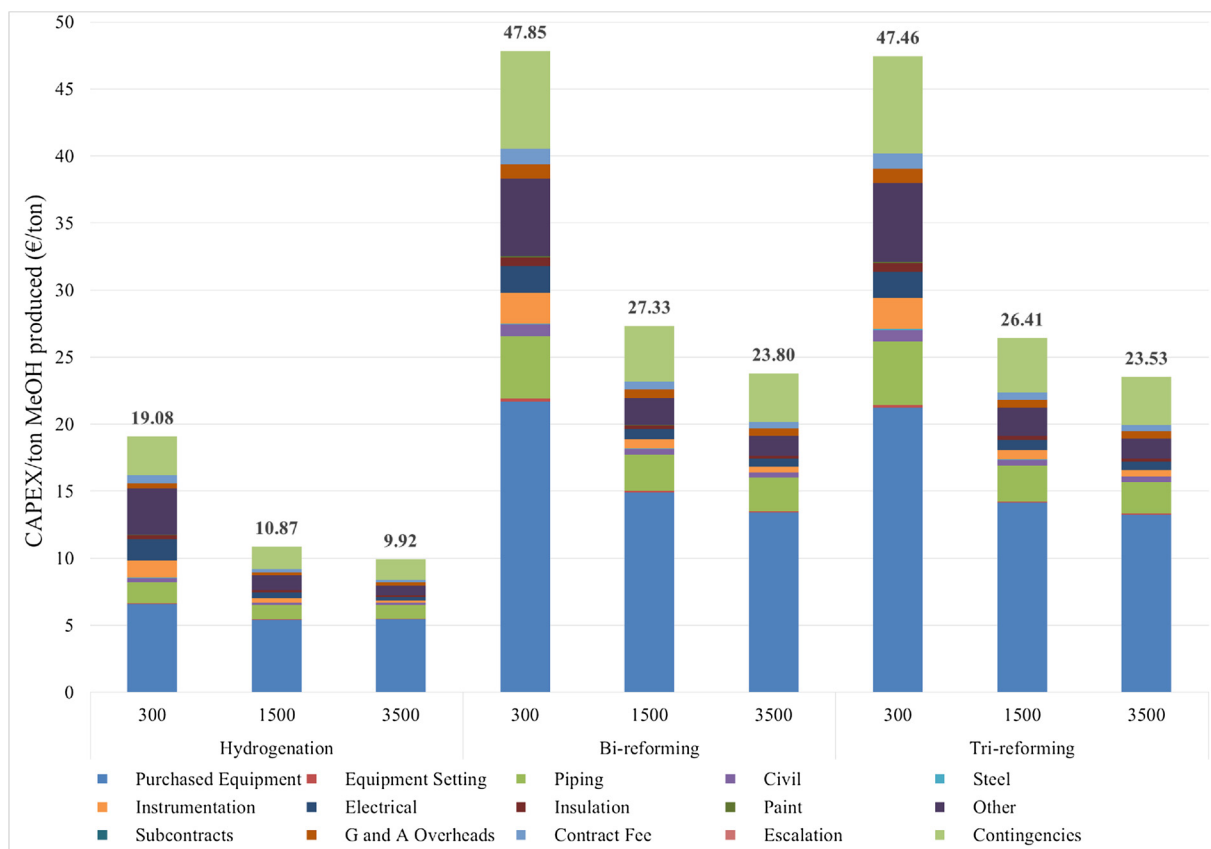


Fig. 6. Annualized capital costs comparisons among different CO₂-based technologies at three methanol capacities (300, 1500, and 3500 ton/day).

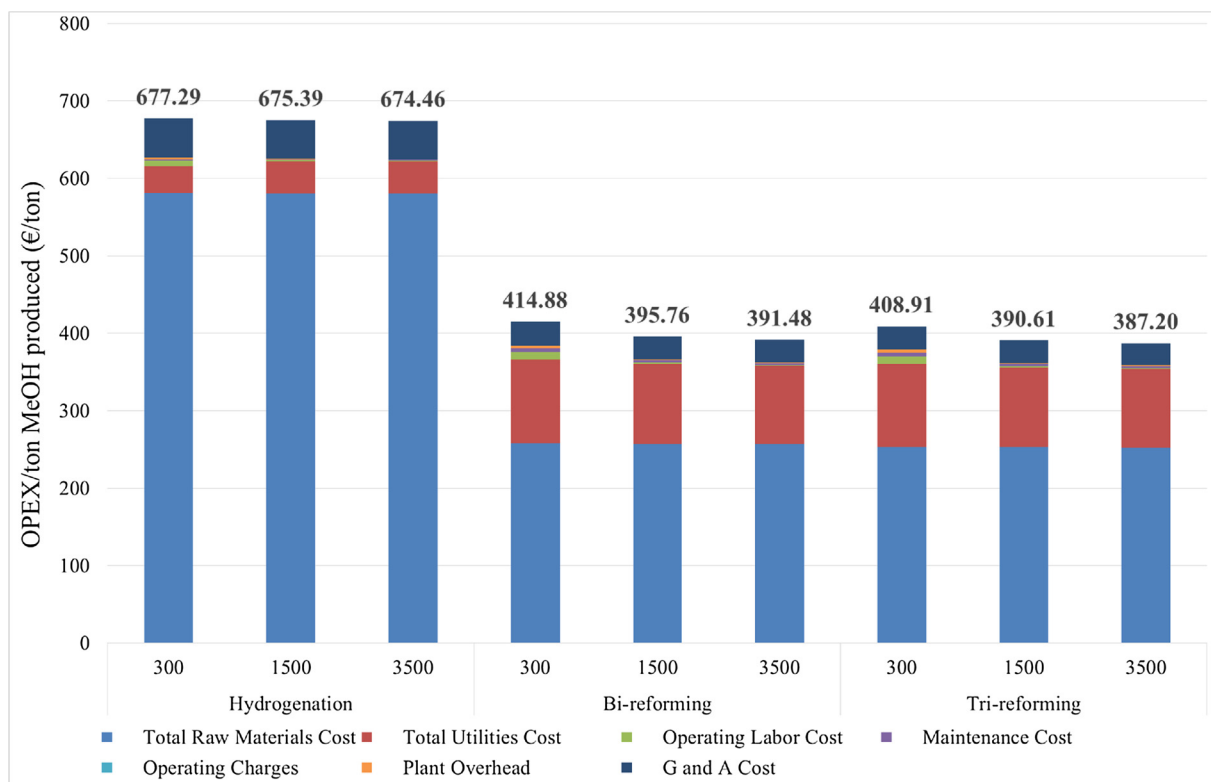


Fig. 7. Annualized operational costs comparisons among different CO₂-based technologies at three methanol capacities (300, 1500, and 3500 ton/day) (hydrogen of hydrogenation processes from steam reforming of natural gas).

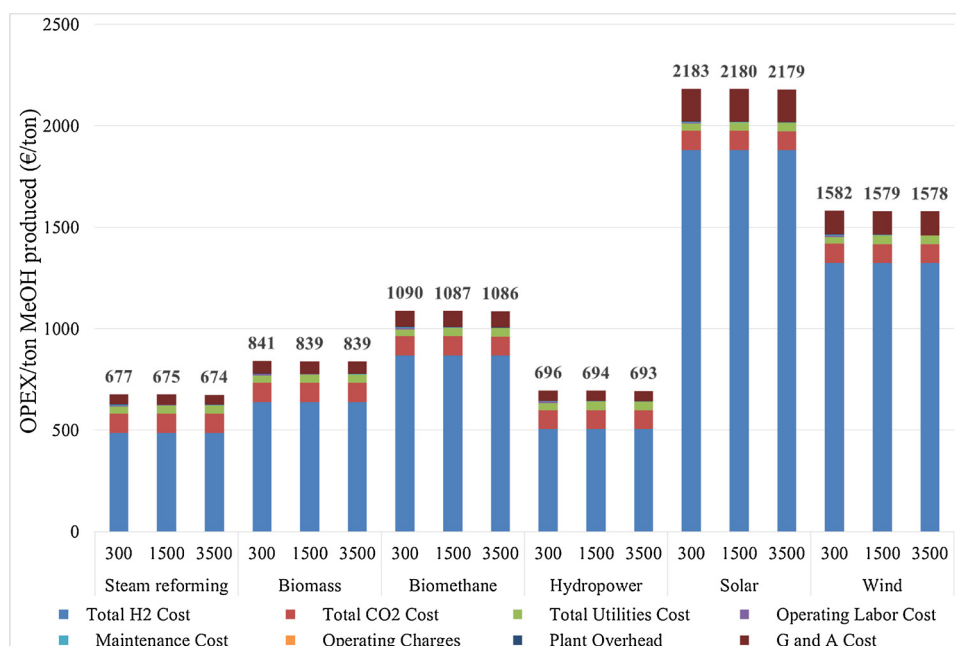


Fig. 8. Annualized operational costs comparisons by using different hydrogen sources at three capacities (300, 1500, and 3500 ton/day).

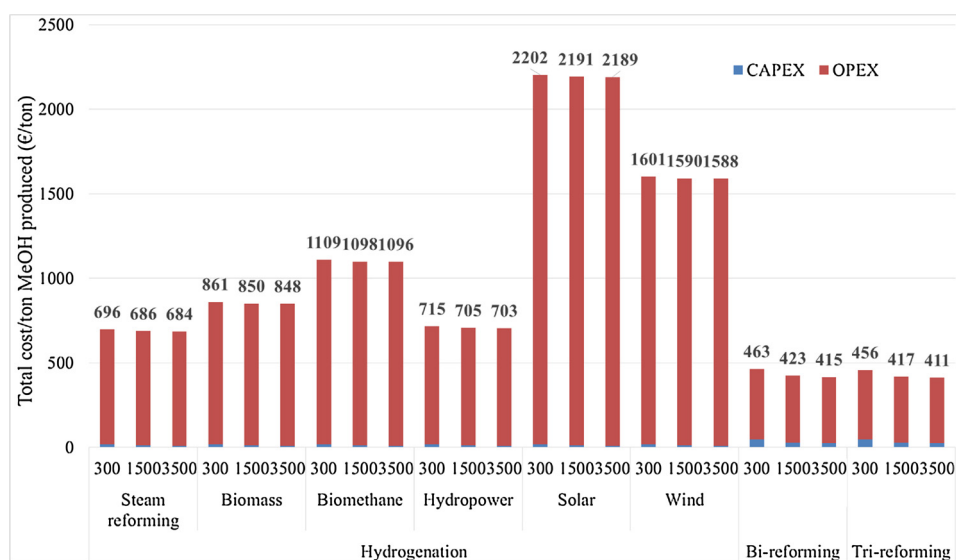


Fig. 9. Annualized total costs comparisons among different CO₂-based technologies at three methanol capacities (300, 1500, and 3500 ton/day) (hydrogen of hydrogenation processes from different renewable sources).

As illustrated in Fig. 6, the investment costs of hydrogenation-based processes are always lower than those of the two other technologies at any capacities. Specifically, at low capacity (300 ton/day), the investment costs for the process using hydrogenation are only 19.08 €/ton while these values are approximately 2.5 times higher, 47.85 and 47.46 €/ton for bi- and tri-reforming respectively. At higher capacities, the costs from these reforming routes slightly decrease; however, still averagely 2.3 and 2.4 times higher than those from the hydrogenation route for 1500 and 3500 ton/day respectively. This is due to the fact that there is a requirement for more processing units (such as the pre-reformer, reformer, etc.) to achieve the desired syngas (with the ratio SR of 2 to 2.1) in both reforming technologies before methanol can be produced. In addition, the investment cost of the CO₂-based process presents a direct relationship to the methanol capacities. The high reduction of 1.8 times of the capital cost can be realized when the methanol production volume increases from 300 to 1500 ton/day for all

scenarios. However, this value is around 1.1 times smaller when the capacity expands from 1500 to 3500 ton/day for all cases. This illustrates the influence of the scale of a process on the methanol production cost. Between reforming technologies, the capital costs from tri-reforming are always lower than those from another technique at any level of capacity. This trend becomes apparent at high methanol production volumes. Clearly, the usage of oxygen allows the reforming to become more efficient, resulting in a reduced equipment size which contributes to the reduction of the capital costs. Another observation from this figure is the cost of purchasing equipment which is above 40% of the total capital costs for all three technologies.

In contrast to the capital costs, the operational costs of hydrogenation-based processes are always larger than those of the routes by reforming as shown in Fig. 7. In particular, at 300 ton/day, the operating cost of the hydrogenation process is 677 €/ton while for reforming-based routes these values are only 415 and 409 €/ton. The gap

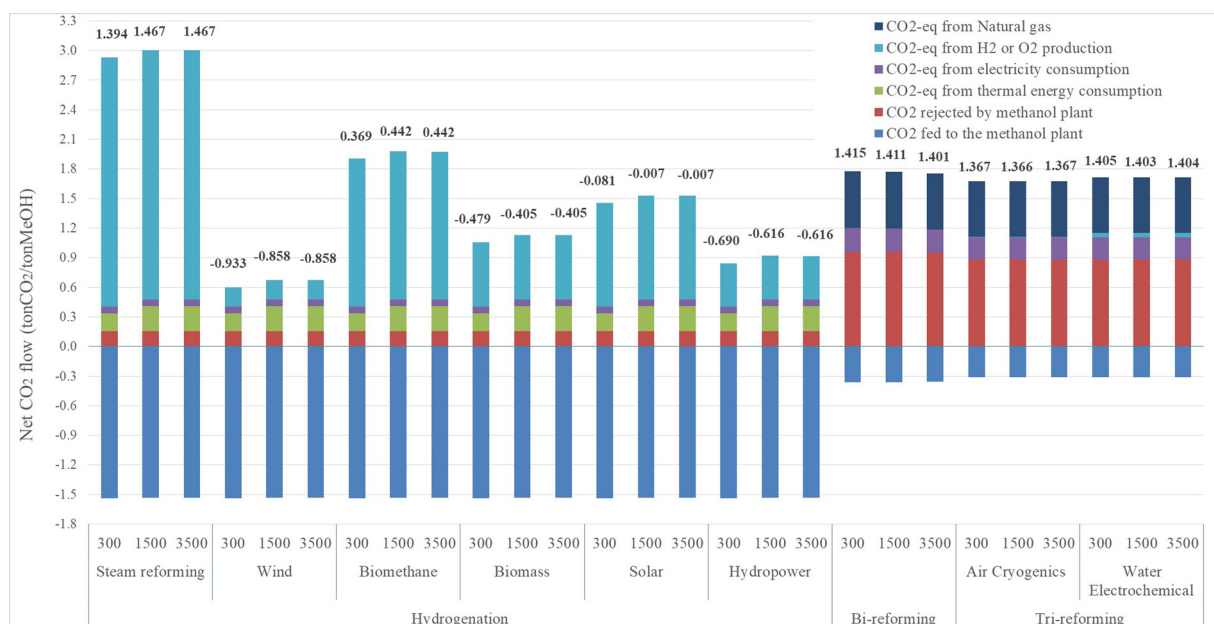


Fig. 10. CO₂ emission components (tonCO₂/tonMeOH) of different CO₂-based techniques at three methanol capacities (300, 1500, and 3500 ton/day). The amounts of CO₂ used by the all processes are shown as negative values.

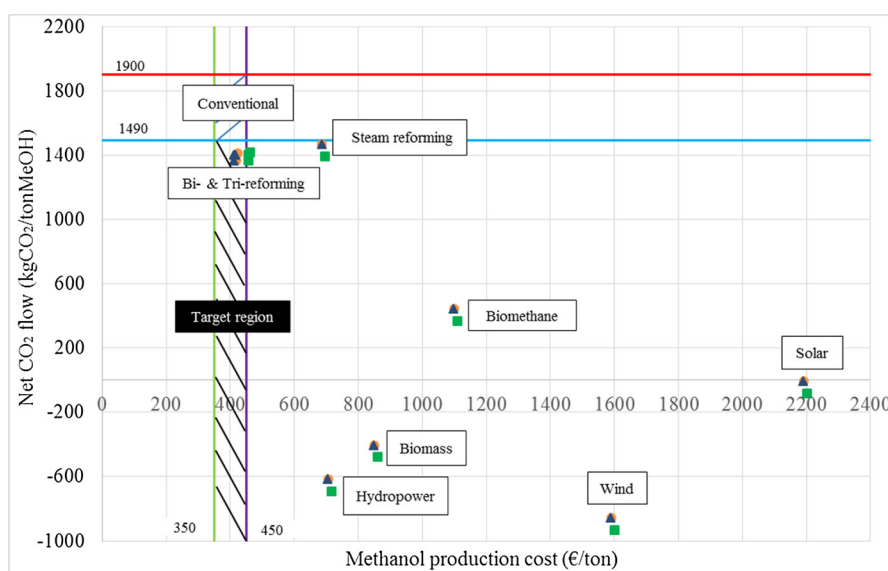


Fig. 11. Net CO₂ flow and the methanol production cost of different CO₂-based techniques at three methanol capacities (300 - green square, 1500 - orange circle, and 3500 ton/day - blue triangle) (hydrogen of hydrogenation processes from various sources).

becomes slightly bigger when the methanol production capacity increases to 1500 and 3500 ton/day, approximately 280 €/ton difference between the hydrogenation and reforming routes. It is clear from Fig. 7 that for any technology the operational costs reduce according to the increase of the methanol plant size. However, this phenomenon is not apparent as shown in the capital costs, especially in the hydrogenation case with only around 3 €/ton difference between small and high capacity. For the reforming processes, the differences are 23 and 22 €/ton for bi- and tri-reforming respectively. Further, it can be seen that the dominant factor for the operating costs for all situations are the raw materials costs, ranging between 61% and 86% for reforming and hydrogenation routes respectively. Different from the hydrogenation process which has a share of utilities costs to the operating cost of only 5%, 25% of the operating cost is accounted for the costs of utilities in the reforming process. This can be explained by the combustion of natural gas to create a high temperature at the reformer.

In the case of hydrogenation, the cost of hydrogen is the most significant contributor to the raw materials costs, resulting in the highest impact on the operating cost. This is attributed to the high purchase cost as well as the high consumption of the amount of hydrogen required to the hydrogenation reactions. As a comparison, the operating costs of using hydrogen from various sources are presented in Fig. 8. As seen from this figure, the costs of buying hydrogen are always much higher those for CO₂, changing from 5 to 20 times higher for steam reforming to solar-based technologies. In general, more than 70% of the operating cost result from the purchase of hydrogen. Moreover, the operating costs of using hydrogen from renewable sources are always higher than those from the steam reforming of natural gas. In specific, the highest cost comes from the solar with high-temperature electrolysis, three times higher than the steam reforming. Only the hydropower energy-based scenarios indicate the approximate values to the steam reforming.

Fig. 9 displays the total costs per ton methanol produced from various technologies and distinct plant capacities. The operating costs make up the biggest part of the total cost for all scenarios. Only around 2 and 10% of the total cost are contributed by the capital costs for hydrogenation and reforming respectively. Moreover, it is evident that the total expenses for reforming routes are always less than those of hydrogenation processes. In particular, for small capacity (300 ton/day), the lowest total cost of hydrogenation-based processes is 696 €/ton, which is 1.5 times higher than that of the tri-reforming route with only 456 €/ton. At higher capacities, the ratio is even higher with 1.6 and 1.7 for 1500 and 3500 ton/day respectively.

It is worth noting that only the methanol total costs from the reforming processes at high capacities are close to the commercial methanol selling price (350–450 €/ton, for the year 2017 [38]). The scenario with the lowest cost of hydrogenation routes (steam reforming of natural gas, 3500 ton/day) costs 1.7 times more than the methanol market cost whereas the most uneconomical case (solar with high-temperature electrolysis, 300 ton/day) is 537% higher above the 2017 market average price level.

8.3. Environmental impact evaluation

From the process simulation results, the direct CO₂ emissions of all cases have been computed. However, the indirect CO₂ emissions are estimated through the values of thermal energy and electricity utilization as well as the amount of H₂ or O₂ employed in all processes. Fig. 10 shows the breakdown of CO₂ emissions (tonCO₂/tonMeOH) for all methods at three different capacities (more detail can be found in Tables S13–S15).

The hydrogenation routes significantly outperform other technologies regarding the reduction of CO₂ emissions if not considering CO₂ emissions from H₂ production as illustrated by Fig. 10. In this case, the consumption of thermal energy and electricity is the dominant reason for CO₂ emissions in hydrogenation technology, 1.6 times higher than that from emissions of direct release. However, when emissions from H₂ production are taken into account, the story becomes entirely different, especially for steam reforming. The average net CO₂ emission for steam reforming changes from -1.08 to 1.44 tonCO₂/tonMeOH, which makes it become the worst alternative among different technologies. Similar to steam reforming, reforming of biomethane from wood has positive values of net CO₂ emission when associated with emissions from H₂ production, approximately 0.42 tonCO₂/tonMeOH. Examination of other scenarios presents that wind power with alkaline electrolysis outperforms others by consuming the most CO₂ than its discharge - around 0.88 tonCO₂/tonMeOH consumption, following by hydropower with water electrolysis and biomass gasification. In contrast, reforming-based routes produce more CO₂ than its CO₂ intake, approximately 1.41 and 1.39 tonCO₂/tonMeOH for bi- and tri-reforming respectively. Between two reforming technologies, the tri-reforming processes always show better CO₂ emission reductions in all scenarios. The lower CO₂ emission of the tri-reforming method is ascribed to the lower energy requirement for heat supply as a result of adding oxygen to reforming reactions. Different from the hydrogenation processes, direct emissions contribute most to the emissions of these reforming technologies, about 53% of total emissions for both routes. Although reforming routes could not consume more CO₂ than their release, their emissions are lower than those of the conventional methanol plant. Specifically, 1.49–1.9 tonCO₂/tonMeOH is released to the air at the typical industrial methanol plant [39–42], which is 1.1 to 1.4 times higher than those from reforming technologies. Obviously, while the CO₂-to-methanol processes based on hydrogenation (with hydrogen from renewable sources) are the long-term target of methanol production from CO₂ and renewable sources-based hydrogen, bi- and tri-reforming routes are one of the propitious transition strategies for the ‘methanol economy’, which not only assist to reduce the exhaustion rate of fossil fuel, but also decrease the dependence on carbon-free hydrogen. For each

scenario, the calculated net CO₂ flow and methanol production cost are summarized and illustrated in Fig. 11. Obviously, there is a trade-off in selecting CO₂ conversion technologies regarding net CO₂ emission and methanol production cost. While reforming-based technologies fulfill economic potential, the hydrogenation route except for hydrogen from steam reforming technology allows a more sustainable solution to produce methanol.

9. Conclusions

In this research, three different routes using CO₂ as feedstock to produce methanol are proposed and developed. All processes are simulated, heat integrated and optimized over three distinct methanol capacities (300, 1500, and 3500 ton/day). A detailed cost calculation involving annualized investment and operational costs are also completed for CO₂-based technologies. Finally, a comparison among various scenarios of three technologies in terms of economy and environmental impact was performed.

This study targeted to investigate the feasibility of employing CO₂ to produce methanol through various routes. The results show that in terms of environmental factors, hydrogenation technology (with hydrogen from renewable sources) is superior to other techniques as a consequence of using more CO₂ than emitting. Therefore, the methanol process from hydrogenation of CO₂ with hydrogen from renewable sources is the best candidate to achieve the near-zero emission goal for the long term. However, bi- and tri-reforming achieve the better results by 37 and 39% lower of the annualized total cost of producing methanol on average, respectively compared to the hydrogenation route. In addition, these reforming options have lower CO₂ emissions than that from the typical methanol plant. For that reason, the reforming technologies can be considered as a transition process from traditional plants to alternative methods because this technology allows a reduction of greenhouse gas emissions but still keeps a strong attraction from the investment. Furthermore, the supply of free hydrogen from industrial by-products or cheaper hydrogen from renewable sources in the future will actively play a part in the decrease of the methanol production cost when hydrogenation route is employed, because it is the central economic element for this cost. With these strategies, the hydrogenation technology can play a vital key in the delay of carbon emissions to the atmosphere.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jcou.2019.05.033>.

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